# Nonlinear Practice 1

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

1. A 143 lb, 21-year-old woman is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 5.1 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 20 mg/L.

**a** (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) a. Calculate the patient's Vmax and KM. What is the KM?

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**b** (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) b. Recommend a daily dose of phenytoin for this patient to achieve a phenytoin plasma concentration of 15 mg/L at steady state.

**c** (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) c. Calculate the half-life and the total body clearance when the plasma concentration is 15 mg/L. What is the half-life?

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**d** (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) d. What will be the steady-state concentration that should be achieved if the dose were 450 mg daily?